

Use of Agile Methodology in System Development

Yasmin Batrisyia binti Zahiruddin¹, Nurul Asyikin binti Khairul Anuar²,
Faculty of Computing,

Universiti Teknologi Malaysia, 8130 Johor Bahru, Johor, Malaysia

¹yasmin.batrisyia@graduate.utm.my

²asyikin04@graduate.utm.my

Abstract—As we know that in the process of software development, we must follow the specific system development life cycle (SDLC). There are many types of SDLC methodologies that can be used in the working environment especially in developing the software. Many researchers have been examined the system development life cycle and has created their own model, which has pros and cons of its own. This analysis paper is focusing on the use of agile methodology in the software development that could plays such a big role in producing a quality product. The paper begins with an explanation of agile methodology. Then, this paper discusses about the comparison between agile methodology and traditional methodology namely waterfall model. It also discusses about the advantage and disadvantages of using agile methodology method in working environments. Lastly, the paper is then wrap up with the conclusion statement.

Keywords—Agile methodology, waterfall methodology, software development, project management, system development life cycle

I. INTRODUCTION

The software development methodology is known as the system development life cycle (SDLC). It is commonly used in IT Technology, several engineering, industry field, etc. The definition of SDLC is a model that used in a project management that has a concept of dividing the main problem into phases. It consists of 6 phases, which are Planning, Analysis, Design, Development, Implementation, and Maintenance. During the system development life cycle, project and program managers have to work align with the system and software engineers, development teams, and end-users. There are many system development life cycle models that has been created nowadays. Each of these models is not perfect. It has its own advantages and disadvantages aspect. These are the few examples of system development life cycle models; Waterfall model, Agile model, Iterative model, Spiral model, V-shaped model, and so on.

The traditional system development model is a Waterfall methodology. Waterfall methodology is a fundamental model in the software development life cycle. Waterfall methodology is a very simple model, but currently it's not been practicing in the industry anymore. This is because, waterfall methodology is quite strict and rigid to the planning system. The meaning of this is, waterfall methodology is easy to use, and it provide a guaranteed output. However, in waterfall methodology, once a phase has been completed, it cannot be changed anymore. And that is why industry nowadays did not use the waterfall methodology because of its less flexibility.

System development has been involving and currently the most popular methodology that been using is Agile methodology. Agile methodology is an interactive methodology where the framework is breaking down to several dynamic phases. It is different from other methodology as it works after each phase, the development team will reflect and check with the clients whether to see if there is anything changes or improvements need to be made as what that they want as the output. Because of its flexibility to make a change, that is why agile methodology is commonly used nowadays. Therefore, this paper will talk more about the use of agile methodology in system development.

II. COMPARATIVE ANALYSIS

A. Differences Of Agile And Waterfall Methodology

In the constantly shifting environment of software development, methods act as guiding principles, shaping how teams approach the creation and delivery of software products. Traditionally, software development was mainly based on sequential methods such as the Waterfall model or related versions. However, as the industry evolved and there was a greater need for flexibility, adaptability, and customer-centricity, Agile methodology gained popularity. Some of the differences of agile methodology and waterfall methodology are briefly described as under:

TABLE I
COMPARATIVE BETWEEN WATERFALL AND AGILE

Methodology	Agile	Waterfall
Roles of Employee	The agile methodology allows team members to work together on project throughout time, resulting in a more self-organizing team structure.[1][2]	Waterfall assigns specified tasks along with responsibilities to each project team member.[1][2]
Communication	The agile methodology Agile values informal communication and encourages regular contact between individuals or small groups of stakeholders.	Waterfall communication is more formal, with specific communication strategies and progress reports distributed to various stakeholders.

TABLE II
COMPARATIVE BETWEEN WATERFALL AND AGILE

Methodology	Agile	Waterfall
Planning	Agile planning is a continuous process that is adjusted as new requirements or information become available throughout the course of the project.[1][2]	Waterfall planning involves a sequential procedure that starts at the beginning of the project and involves a detailed outline of all criteria and goals.
Speed	Agile projects are typically completed faster than waterfall projects due to the iterative development cycles employed in agile.	Waterfall projects typically take longer since all needs must be agreed upon before development can begin.
Documentation	Agile relies on minimum documentation and emphasizes self-organizing teams and cooperation.	In order to make sure that every team member is aware of the process, waterfall development mostly relies on meticulous documentation of every stage.

III. ADVANTAGES AND DISADVANTAGES OF AGILE METHODOLOGY

A. Advantages

- Along the progressing phases, agile methodology could **provide flexibility to change the requirements in a short time** as what the client needs before proceeding into the next phases. This is because the focus is on the brief iterations only that typically will last between one to two months.[7] Therefore, developers shouldn't against changes instead welcoming the changing requirements, and learn to take advantage of it.

- Agile methodology could **create a good teamwork spirit** inside the working society. This is because in order for the project to be successful, it's not only one person working but it has to include a lot of people. [11] For example, developers and business people have to work closely, so that, they could share their knowledge and learn from each other and achieve higher working productivity. [12]

- Throughout the process of fulfilling with the project scope, using Agile methodology could make the **process of identifying errors more quickly**. This is because it is based on the short feedback-and-development cycles. Therefore, it could save a lot of time and cost for the developers and clients as the errors will be directly corrected. [8]

- Having a frequent meeting and discussion within the development team and the customers, it could **increase the improvement of developing the good product and makes the customers satisfied with the work** especially when it is totally done [e-book] Not only that, having a frequent meeting also could avoid from having a miscommunication opportunities between them. [11]

- Using a roadmap would help the team in **reflecting their learnings** and, if required, modifying their development method. By applying this, the team may be able move forward, accomplishing their objectives and generating the greatest possible output. [11]

B. Disadvantages

- Too much of flexibility also could effect level of predictability to be lower**. This is because it is hard to estimate the time, or quantify the resources and effort to be done accurately. Therefore, many teams fear of this project uncertainty because it would give a big impact to the company as they would feels frustration and make a poor decision making [11].

- Many businesses believe that in order to achieve a good results using an Agile Methodology, it can only be achieved by requiring a team that filled with **successful and talented individuals only** who favor in many degrees as they have to make the changes requirements fast [9].

- Using an agile methodology, the documentation is not really necessary as it is on the back burner, and that is what makes the **documentation is questionable**. This is quite a big disadvantage of what developers could produce because the documentation is an important part of delivering a good product. This is because it could helps in a process of maintenance. For example, whenever a new worker come in, they would know how to handle the product because it has been documented properly [10].

CONCLUSION

To sum up, the contrast between the Agile and Waterfall approaches emphasizes how software development is moving toward a more adaptable and customer-focused methodology. Moreover, Agile is notable for its flexibility and iterative structure, which enables teams to effectively adapt to evolving requirements. Agile's iterative cycles encourage regular client interaction, which ensures that the product is in line with customer needs. Despite issues like as documentation and predictability, Agile's emphasis on collaboration and gradual progress has earned it widespread acceptance in company culture. Agile methodology continues to be an important component of current software projects, fostering efficiency, creativity, and client satisfaction as software development continues to change. The implementation of it represents not just a procedural shift, but also a cultural shift toward accepting change, client collaboration, and other factors for managing the continually shifting environment of modern software development.

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